Lecture 02

PROJECT THESIS OUTLINE
(Chapter 1 – 5)

CSP650

Writing and Structuring Report



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CSP 650 (5 chapters)

Chapter 1

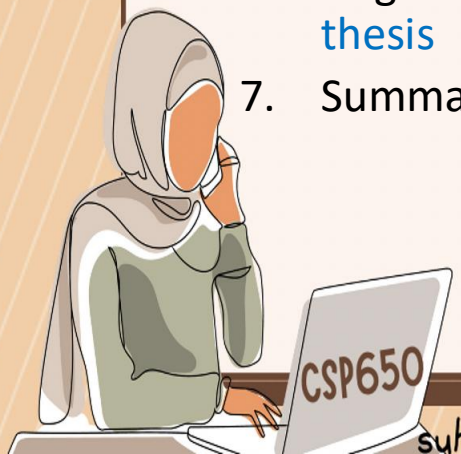
1. Background of Study–Current Issues and motivation
2. Problem Statement
3. Aim
4. Objectives
5. Significances
6. Report Structure :organization of thesis
7. Summary

Chapter 2

1. Overview of research domain
2. Overview of current trend/technology used
3. Comparison of techniques/methods
4. Related works/existing similar systems
5. Justification of selected technique/methods in discussion
6. Summary

Chapter 3

1. Overview of methodology
2. Description of methodology phases
3. Flowchart/diagram of systems development
4. Implementation of technique used
5. Hardware/software implemented
6. Summary



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CSP 650 (5 chapters)

Chapter 4

1. Overview of findings and analysis
2. Result of Obj 1
3. Result of Obj 2
4. Result of Obj 3
5. Diagram/chart representing analysis
6. Justification of results obtained
7. Summary

Chapter 5

1. Overall conclusion (objective by objective's achieved)
2. Systems Strength
3. Systems Limitation
4. Future enhancement (recommend to overcome the limitation)



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CSP 650 (SRS and SDD)

SRS

1. Follow SRS template
2. Modify according to systems (map with your own system)

SDD

1. Follow SDD template
2. Modify according to systems (map with your own system)



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UPDATED CHAPTER 1



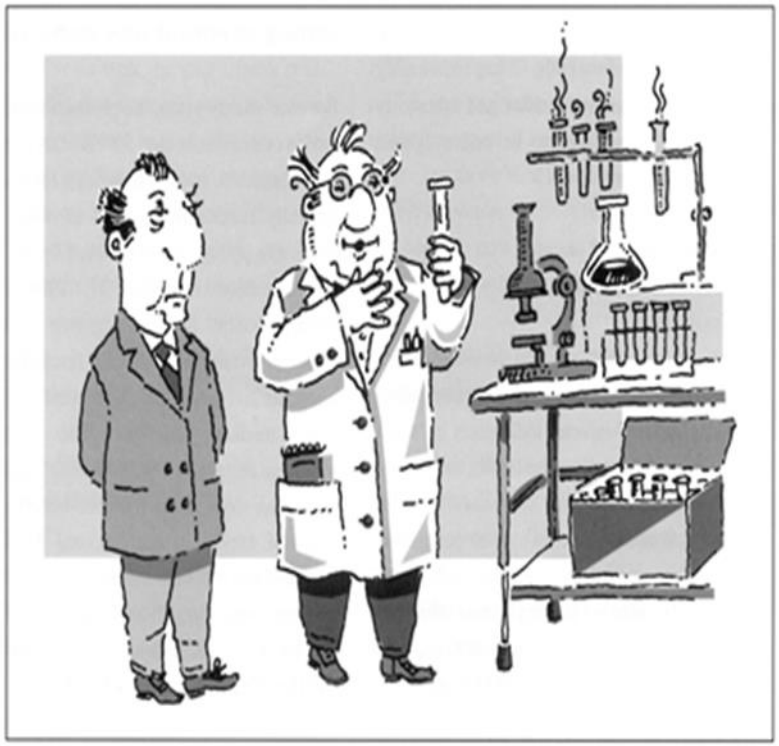
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Revise your word proposal -> Thesis

Course Learning Outcome 1:
Solve undertaken problem using appropriate technique and principles

Course Learning Outcome 2:
Produce project report with good analysis and proven result in appropriate format

Course Learning Outcome 3:
Demonstrate an understanding of project through oral presentation



"While it's not a cure, it does mean a guaranteed income for me."

Why you must do it?

Demonstration of an understanding of the state of your work.



Rule of Three

- Within each chapter, repeat yourself 3 times
  - Intro. ***This thesis*** *will show ..*
  - Body. *Show them ..*
  - Concl. (summary) ***This project*** *have shown ..*

- Within report, repeat your contributions 3 times
  - Intro chapter (CHAPTER 1) – in significances
  - Main chapters (CHAPTER 4) – in result analysis
  - Conclusion chapter (CHAPTER 5) - introduction

- But don't bore reader
  - E.g. in introduction be brief, in conclusions be broader



Parts of Your Report

- THREE parts
  - Preliminaries
  - Body of the text
  - References

- Acknowledgements
- Acronyms
- Figures
- Tables
- Declarations
- etc

- Use APA style



Body of Your Report

- **THREE** parts with **FIVE** major sections:
  - Introduction
  - Body of Report
    - Review Of Related Literature
    - Methodology
    - Results And Discussion
  - Summary, Conclusion And Recommendation



BODY OF Your Report

- The body of the thesis report contains five logical divisions :

1.Introduction → **2. Literature Study** → **3. Methodology** → **4. Results and Analysis of Data** → **5. Conclusions and Recommendations**

NOTE: please always rephrase and write your source in **any sections** if you mention an idea which is not yours.



Overview of Introduction Chapter

- **Find problem in related works?**
- **Write a brief account on the current state of the problems.**
- **Explain what you going to solve and your objectives.**
- **Explain your scope / limitation.**
- **End your Introduction by writing a brief information on how your thesis is organized**
- **Add new statistic/survey findings in INTRO**



Background, Definition of Terms

- An overview of the overall topic.
- Current Issues (add latest references)
- What is already known? This will begin to lead us to your Review of Literature (Chapter 2).
- Definition of Terms (If there's new terms, modify whole report)



CONVERT to PAST TENSE

- **Proposal -> Report/Thesis**
- **Past tense**

….. This thesis entitled XXXXX elaborated …

….. The aim for this project is…

….. This system ~~will~~ provides significances…



Problem Statement,
Aim,
Objectives

- **Write a brief account on the current state of the problems.**
- **Explain what you are going to solve and your objectives. (modify if there's any changes after proposal defense)**



ENSURE PROBLEMS tally with AIM

Problem Statement:

There is no appropriate tool to provide control in managing project artifacts for students in Bachelor of Information Technology (Information Systems Engineering) undertaking 3 project-based core courses (ITS470, ITS570, ITS670) in 3 consecutive semesters where the deliverables of one course are input to the other.

AIM:

To develop a system to manage ISE Students' artifacts using cloud computing technology for team members to collaborate, standardize, track and share their work and for lecturers to monitor the progress and control the students' project progress - Collabtifacts, which combines *Collab*oration and ar*tifacts*.



Objectives

- Breaking the project into **milestones**:
  - To gather the requirement of Clinic ABC and analyze symptoms of Dengue Patients
  - To design the Dengue Diagnostic Expert System by using Rule-based Technique
  - To develop the system of Dengue Diagnostic Expert System



ScopeS

- **Explain your scope /boundaries.**
- **Data used (range for year 2015 until 2024)**
- **Target user are employees of XXX**
- **The system did not cover payment gateway**
- **This mobile apps developed by using hybrid platform.**



Scope

- The extent of the area or subject matter that you are solving in your project. Boundary. Population of study, the instrument, data gathering method, previous knowledge, etc.

Example:

- This development project focus on students and lecturers in Bachelor of Information Technology (Hons.) Information Systems Engineering from Faculty of Computer and Mathematical Sciences, UiTM Jasin.



SignificanT

Recipients who benefit from your project

Example:

- Able to manage patients data efficiently and effectively
- The cost of storage management can be saved through the practice of cloud computing.
- Reduce time consuming to diagnose the dengue fever level for various symptoms
- Patients data and privacy well managed



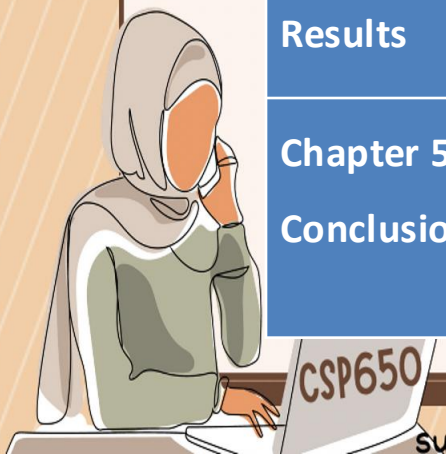
Report Structure

- **End your Introduction by writing a brief information on how your thesis is organized**



Report Structure

Chapter Title	Description
Chapter 2: Literature Review	This chapter consists of literature reviews from various sources on cloud technology and how it is used for storage management.
Chapter 3: Methodology	Chapter 3 expresses the methods used and phases that this research covered. As well as the hardware and software requirements to develop and use Collabtifacts system.
Chapter 4: Findings and Results	This chapter shows the results of work done for each phase described in the methodology chapter. Thorough analysis and discussions of each findings are described here.
Chapter 5: Conclusion	This chapter concludes the research with discussion of the benefits gained with the system and recommendation for future works.



Summary

- a brief statement or account of the main points of what you have discussed above.
  - Background
  - Current issues
  - Your aim
  - Significant
  - Scope



UPDATED CHAPTER 2



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Updating your literature content

- Rationale for engaging in primary research/project
- Critically analyze a segment of a published body of knowledge through summary, classification, and comparison of prior research studies, reviews of literature, and theoretical articles
- Introduction, body & conclusion focus on your domain



LR - Introduction

- Define or identify the general topic, issue, or area of concern
- Point out overall trends in what has been published about the topic
- Establish the writer's reason (point of view) for reviewing the literature
- Explain the criteria to be used in analyzing and comparing literature
- Organization of the review (sequence)
- State why certain literature is or is not included (scope)



LR – Body

- Group research studies and other types of literature
- Summarize individual studies or articles



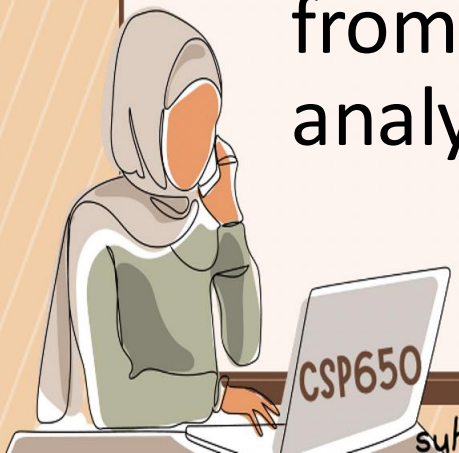
LR - Conclusion

- Summarize major contributions of significant studies
- Evaluate the current "state of the art"



Revised LR

- Add references up to 30 (those with no dates (n.d.) not counted.
- Add latest references regarding your domain
- Add a paragraph of explaining your comparison and justify your reasoning.
- DO NOT copy methodology references from senior report – will increase original analysis value and resulted in plagiarism.



Reference

- http://writing.wisc.edu/Handbook/ReviewofLiterature.html
- APA styles version 7



UPDATED CHAPTER 3



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RECAP : What is Methodology

- Approach
- Methods
- Design
- Deliverables

Describe **what** you did, **why** you did it, **how** you did it, **what** tools/techniques did you use, **what** did you deliver, **which** objective did you achieved, and **when**



Methodology TABLE (Example)

Phase	Activity	Deliverable	Objective
Phase 1	Activity 1.1 Activity 1.2 : : Activity 1.x	Deliverable 1 : : Deliverable x	Objective 1
Phase 2	Activity 2.1 Activity 2.2 : : Activity 2.x	Deliverable x+1 : : Deliverable y	Objective 2
Phase 3	Activity 3.1 Activity 3.2 : : Activity 3.x	Deliverable y+1 : : Deliverable z	Objective 3



- This phase involved the finding of requirements for the project. The system scope, system functionality, assumptions and depen[illegible] figured out. System overall process are documented in this phase. Unified Modeling Language (UML) standard of software engineering modeling was used in the system. Requirements are gathered in two phases: a preliminary [illegible] questionnaire on ISE students to get their problems, an interview with a student from ISE to get an in-depth details of their problem. From the requirement gathered, the problem domain were descri[illegible] Use Cases, which relates the system functional requirements. The overall architecture model was also developed t[illegible]mentation placement of the system from the given re[illegible]udent. The preliminary system hardware and softwar[illegible] reviewed in this phase and edited according to the current requirement. Domain classes of user, and data objects of th[illegible]ed in UML Class Diagram where the name of the class, [illegible]of each object are defined. UML Activity Diagram and UML System Sequence Diagram that explain each flow of use cases interaction between the user and the system was developed. These details gathered are concluded in a Software Requirements Specification (SRS) which are compiled in the appendix A of this project. At the end of this phase Objective 2 was achieved.

Why you did it

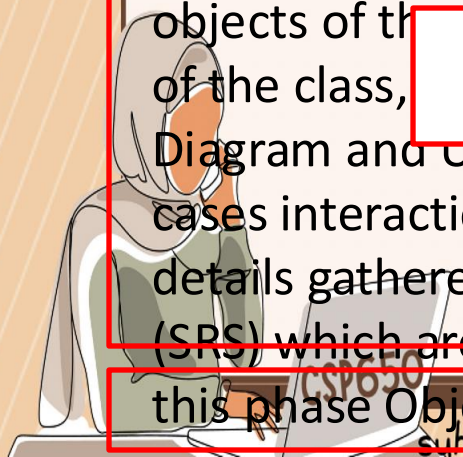
What you did

How you did it

What tools and techniques did you use

What was delivered

Which objective



Note

- Please do not put your problem identification as part of your methodology – that was solely for your proposal (exclude deliverables of problems identified, interview conducted)
- Phase of data requirement and analysis can be combined and directly elaborate on the analysis deliverables.
- Revise the hardware and software requirements based on your developments.

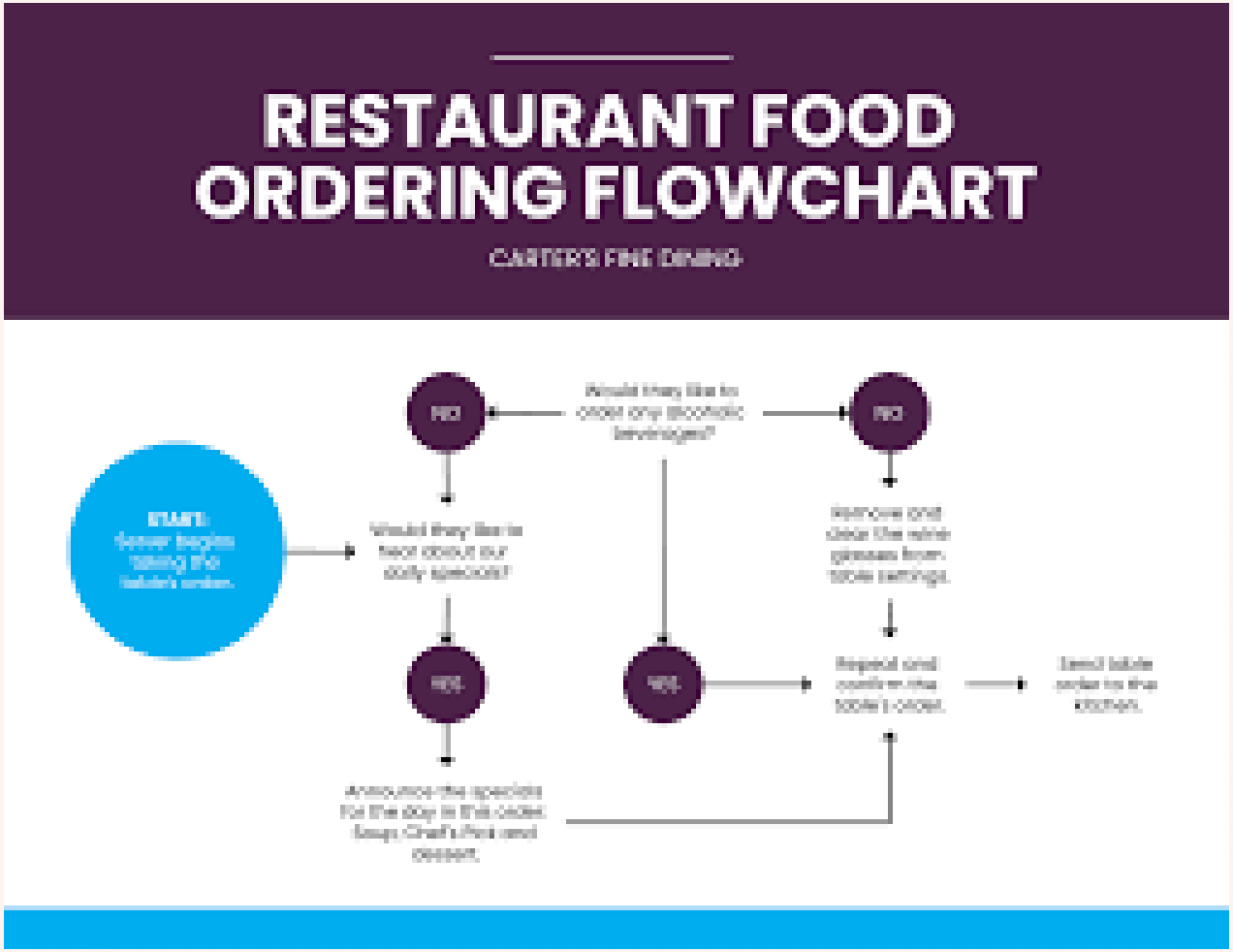


Revised Methodology

1. Overview of methodology
2. Description of methodology phases
3. Flowchart/diagram of systems method (optional)
4. Implementation of technique used (eg: rule-based architecture) - optional
5. Hardware/software requirements
6. ~~Milestone/Gantt chart~~ (REMOVED)
7. Summary

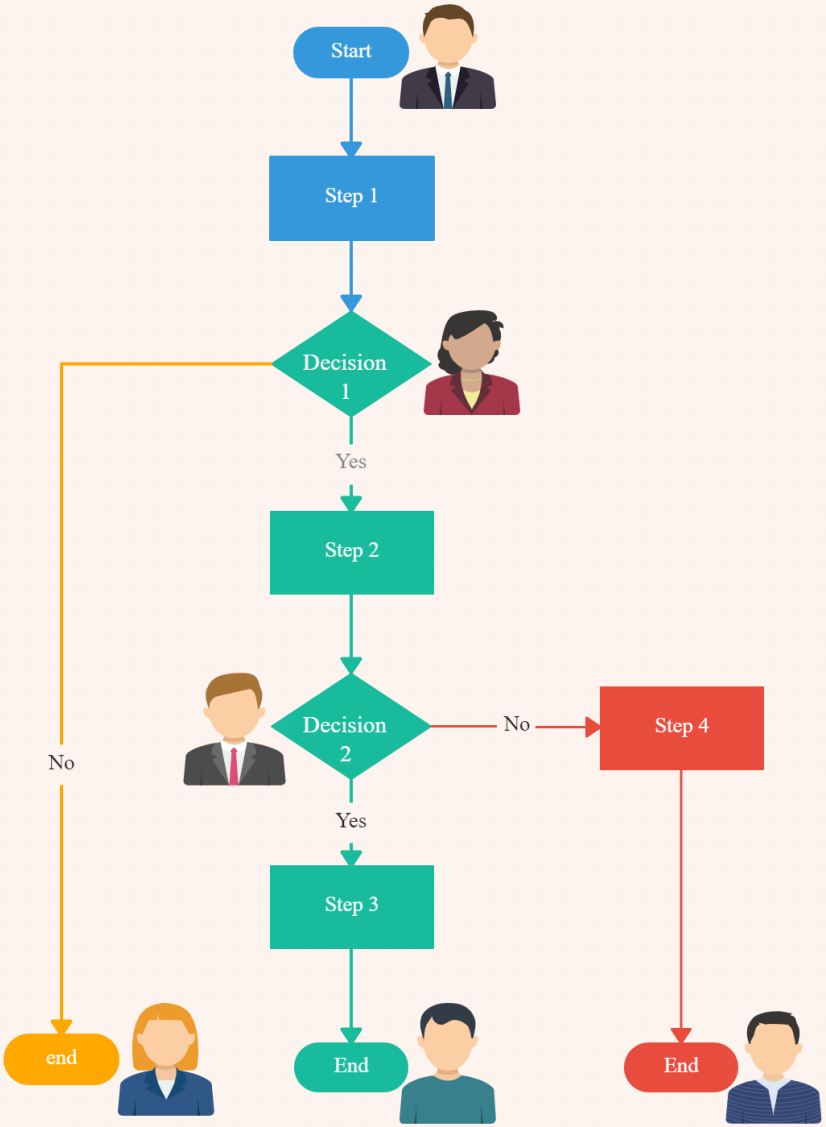


Flowchart example





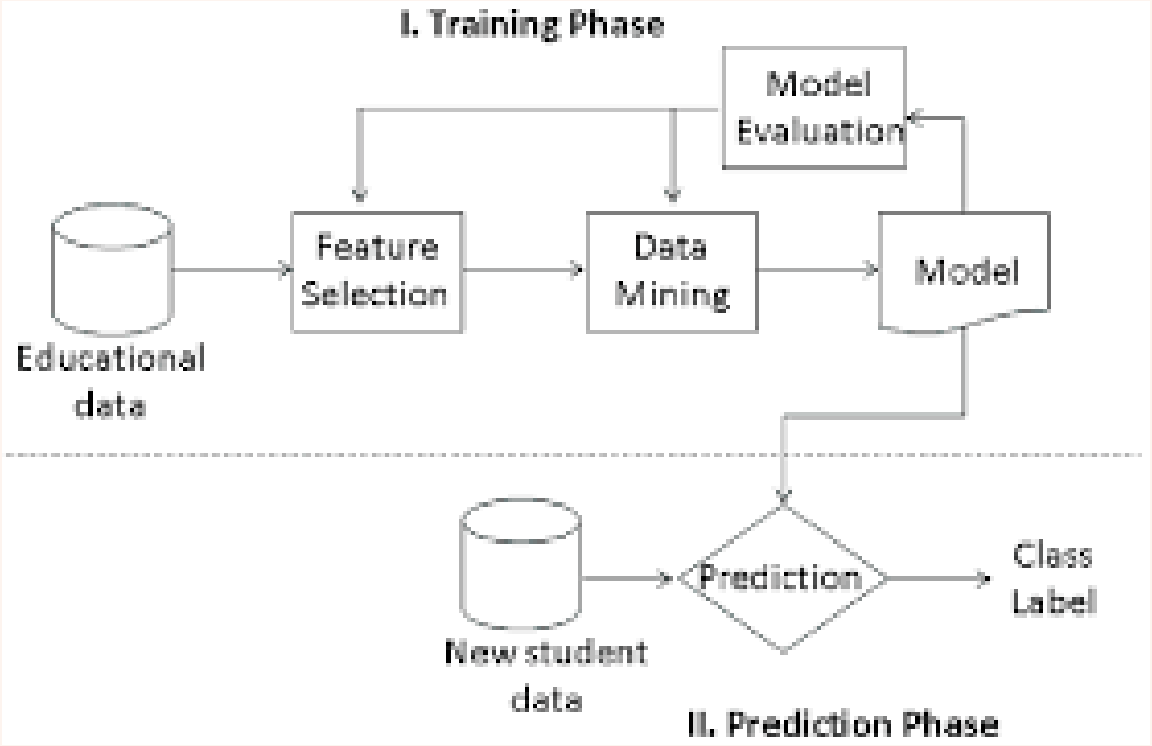
Flowchart example





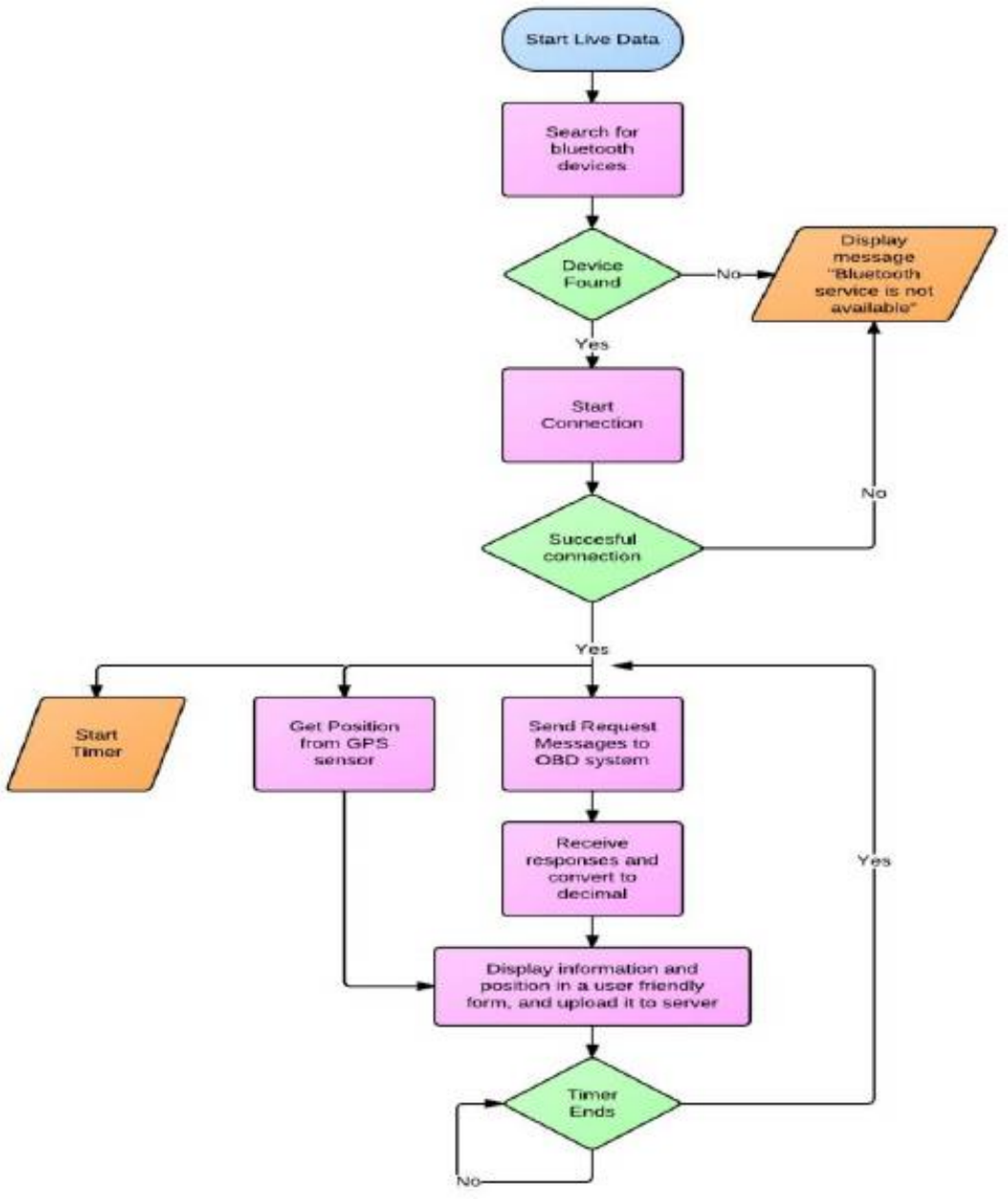
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Flowchart example





Flowchart example





Implementation of technique used (example)





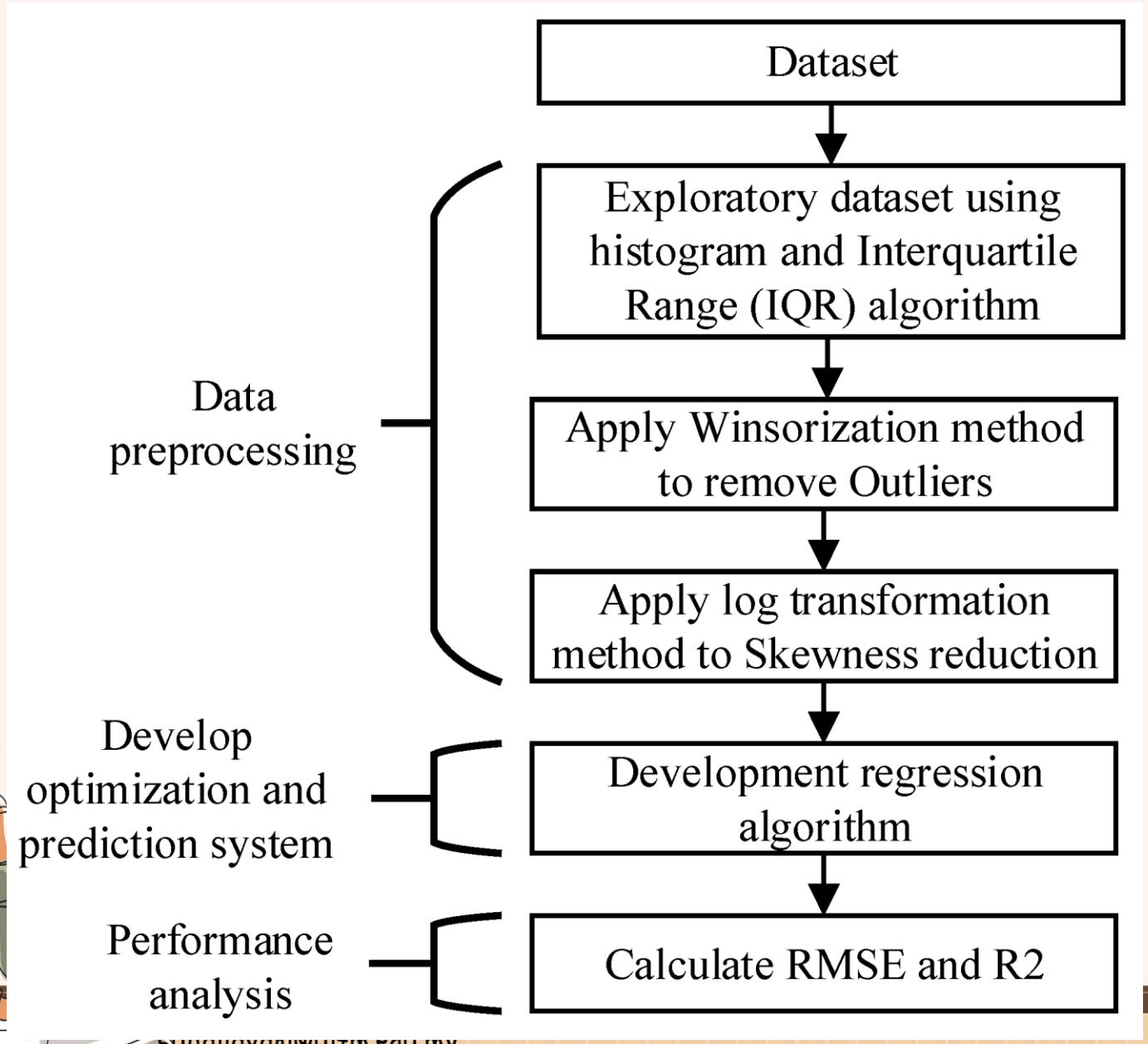
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Implementation of technique used (example)



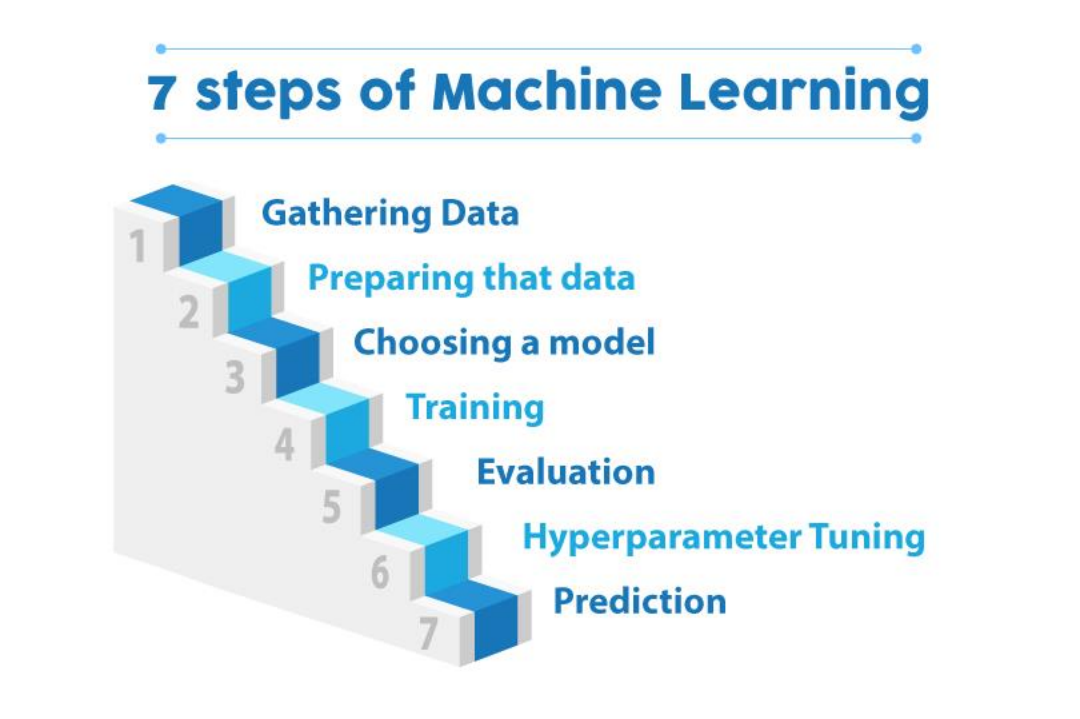


Implementation of technique used





Implementation of technique used(example)





Pages Breakdown

Chapter	Weight	Percentage of Num of Pages	Eg (Total pages 50)
Chapter 1 – Introduction	2	12.5%	**~ 5**
Chapter 2 – Literature Review	3	20%	**~ 15**
Chapter 3 – Methodology	4	25%	**~ 27**
Chapter 4 – Results/Findings/Prototype	5	30%	
Chapter 5 – Conclusion and Recommendation	2	12.5%	**~ 3**



Structure of the Thesis

1. Cover
2. Title Page
3. Supervisor approval
4. Student's declaration
5. Acknowledgement
6. Abstract
7. Table of content
8. List of figures
9. List of tables
10. List of abbreviations
11. Chapter 1 – 5
12. References
13. Appendix A : CLM
14. Appendix B : SRS
15. Appendix C : SDD
16. Appendix D : Data
17. Appendix E : Source code
18. Appendix F : Stakeholder approval



Refer to FYP guide book, appendix M

- Before submit report to supervisor for review, fulfill this checklist
- Submit this checklist to your supervisor together with the report

ADDITIONAL FOR CS266 STUDENTS:
APPENDIX : SRS & SDD - COMPULSORY



FYP Report checklist

Please ensure that you have all items listed below in your report and tick the box.

	ITEMS	REMARKS	PAGING	'✓' the box when finished
1	Cover Page	compulsory	none	□
2	Title Page	compulsory	i *(but not printed/hidden)*	□
3	Supervisor Approval	compulsory	ii *(but not printed/hidden)*	□
4	Student Declaration	compulsory	iii *(but not printed/hidden)*	□
5	Acknowledgement	compulsory	iv	□
6	Abstract	compulsory	v	□
7	Table of Contents	compulsory	vi	□
8	List of Figures	compulsory IF the report has figures or diagrams	continue from the previous page number	□
9	List of Tables	compulsory IF the report has tables	continue from the previous page number	□
10	List of Abbreviations *OR* Glossary	optional	continue from the previous page number	□
11	Chapter 1 : Chapter 5	compulsory	1 : continue from the previous page number	□
12	References	compulsory	continue from the previous page number	□
13	Appendix A : Appendix X **SRS SDD**	optional **compulsory**	continue from the previous page number	□
14	Blank Paper	compulsory	none	□

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THANK YOU !

See you in next Lecture!



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